Course Name: Geographic Information System

Week 09 - Assignment 09 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total marks: 10x1= 10**

QUESTION 1

Which of the following is a logical operation instead of an arithmetic operation?

a) Addition
b) Multiplication
c) Equal to
d) Subtraction

Answer: (c)

Detailed Solution:

Equal to is a logical comparison operator, while addition, subtraction, and multiplication are arithmetic operations.

QUESTION 2

The result of a buffer operation in GIS is generally a _____ feature.

a) Point
b) Polygon
c) Raster
d) Line

Answer: (b)

Detailed Solution:

Buffering creates polygon zones around spatial features at a specified distance.

QUESTION 3

Network analysis in GIS mainly depends on _____.

a) Connected networks
b) Movement rules
c) Units of measurement

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d) All of the above

Answer: (d)

Detailed Solution:

Network analysis requires connected paths, movement rules, and measurement definitions for routing and accessibility analysis.

QUESTION 4

Bilinear interpolation uses the nearest _____ neighboring pixels.

a) 4
b) 9
c) 16
d) 25

Answer: (a)

Detailed Solution:

Bilinear interpolation calculates new pixel values using the 4 nearest neighboring pixels.

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QUESTION 5

Bicubic interpolation calculates output values using _____ neighboring pixels.

a) 4
b) 8
c) 16
d) 25

Answer: (c)

Detailed Solution:

Bicubic interpolation considers the nearest 16 pixels to produce smoother image resampling results.

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QUESTION 6

A GIS model representing terrain as continuous non-overlapping triangles is called _____.

a) DEM
b) Contour
c) TIN
d) Spline

Answer: (c)

Detailed Solution:

TIN (Triangulated Irregular Network) represents surfaces using connected triangular facets.

QUESTION 7

A line joining locations of equal elevation on a map is known as a _____.

a) Spline
b) Contour
c) Aspect line
d) Ridge line

Answer: (b)

Detailed Solution:

Contour lines connect points having the same elevation above mean sea level.

QUESTION 8

The rate of elevation change over a surface is called _____.

a) Aspect
b) Slope
c) Relief
d) Curvature

Answer: (b)

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Detailed Solution:

Slope measures the steepness or rate of change in elevation between locations.

QUESTION 9

Overlay analysis in GIS can be applied on _____.

a) Raster data only
b) Vector data only
c) Both raster and vector data
d) Neither raster nor vector data

Answer: (c)

Detailed Solution:

Overlay operations combine multiple spatial layers and can be performed on both raster and vector datasets.

QUESTION 10

In remote sensing, sensors that emit their own energy to capture data are called _____ sensors.

a) Passive
b) Active
c) Optical
d) Thermal

Answer: (b)

Detailed Solution:

Active remote sensing systems, such as radar, transmit their own energy and record the reflected signal from Earth's surface.

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